DATATHON 2026: Strategic Revenue Audit of the Vietnamese Fashion E-commerce Market (2012–2022)

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Abstract

This report details a strategic audit of a decade-long (2012–2022) longitudinal dataset from a leading Vietnamese fashion e-commerce enterprise. The business has achieved massive scale, with a cumulative revenue of **16.43 billion VND** across 650,000 orders. However, this audit reveals that top-line growth is masking severe operational leaks. Using forensic data linking, we expose a collapse in customer retention, a loss-leader strategy that destroys basket profit, an inelastic logistics response that wastes shipping spend, and a predictable seasonal demand shock that strains inventory. We conclude with immediate prescriptive actions to plug these leaks, prioritizing margin discipline and loyalty over simple acquisition volume.

1 Part 2: Strategic Forensic Audit

This audit synthesizes four major research directions (Directions 01–04) into a unified business diagnosis: top-line revenue is masking severe structural leaks in retention, pricing, logistics, and seasonality. Together they constitute a "leaky bucket" — the growth engine has demand, but loses value at every stage after acquisition.

Leak	Direction	Evidence	Prescriptive Action
Loyalty Collapse	02	Retention 64.7%→6.7%	Lifecycle CRM & loyalty tiers
Loss-Leader Trap	01	Profit -1,473 VND/order	Enforce bundling & margin floor
Inelastic Ship.	03	74% repeat regardless of speed	Downgrade to economy shipping
Campaign Paradox	04	Year-End: -46% rev, 1.5% margin	Tiered rewards, hard margin floor

Table 1: The Four Structural Leaks across research directions.

1.1 1. Loyalty Collapse — Direction 02 (Customer Lifecycle)

Our Direction 02 Customer Lifecycle analysis computed next-year retention for every acquisition cohort from 2012 to 2021 by joining `orders`, `customers`, and `order_items` across order-date windows. **Descriptive:** The 2012 cohort retained 64.7% of customers the following year; the 2021 cohort retained only 6.7%. **Diagnostic:** The business has shifted from a community-driven brand to a transactional storefront. The acceleration after 2018 — confirmed in Figure 7 (Appendix B) — aligns with the Double-Day discount era, which attracted low-LTV bargain hunters. **Predictive:** If the trend holds, Customer Acquisition Cost (CAC) will exceed LTV within two years. **Prescriptive:** Freeze broad acquisition spend; redirect to lifecycle CRM, post-purchase reactivation flows, and tiered loyalty rewards.

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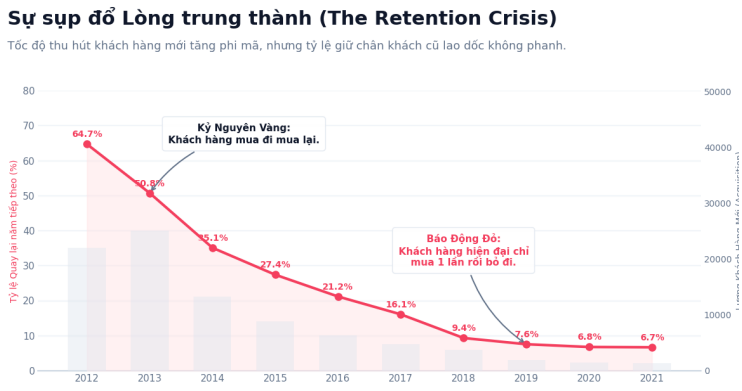

Figure 1: Next-year retention across annual customer cohorts, 2012–2021 (Direction 02).

1.2 2. Loss-Leader Trap — Direction 01 (Product Dominance)

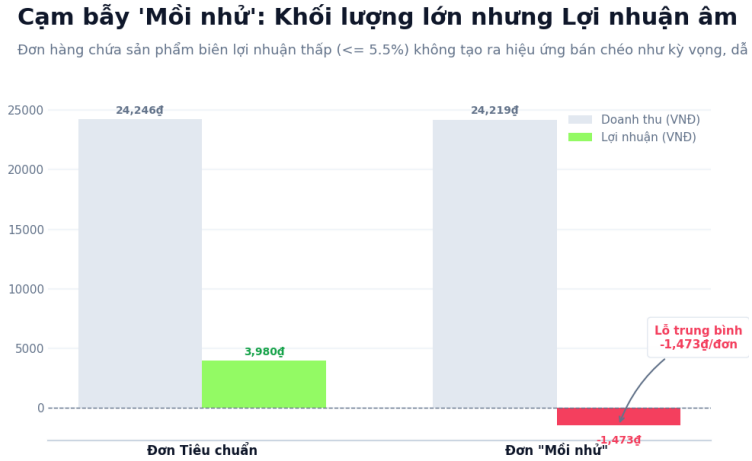

Figure 2: Basket economics: loss-leader orders vs. standard orders (Direction 01).

Direction 01 audited every SKU's unit margin using `products.cogs` and `products.price`, then linked to `order_items` to measure downstream basket effects. **Descriptive:** 461 SKUs operate at $\leq$5.5% gross margin; orders containing them average $\sim$24.2K VND revenue — but basket profit collapses to -1,473 VND versus +3,980 VND for standard orders. **Diagnostic:** Cross-sell is failing: customers add only +0.22 extra items per basket. The strategy loses money without generating the compensating upsell it was designed to trigger. Figure 5 (Appendix A) further shows that L/XL sizes carry the highest margin per unit — the assortment mix is skewed against the company's own best pockets. **Predictive:** Continued loss-leader reliance will compress gross margin irreparably. **Prescriptive:** Discontinue standalone loss-leader selling; gate purchases behind a basket-margin floor or hard bundle.

1.3 3. Inelastic Logistics Paradox — Direction 03 (Operational Friction)

Direction 03 linked `shipments`, `orders`, and repeat-purchase flags derived from `order_items` to test whether delivery speed predicts customer return. **Descriptive:** The repurchase rate holds flat at $\sim$74% across the full 2–14 day delivery spectrum. **Diagnostic:** Brand loyalty, not logistics, drives repurchase; premium 3PL costs are wasted. The same direction's seasonal audit — see Figure 8 (Appendix C) — shows Tet demand surges 40–80% at T-30, then collapses, creating avoidable stockout stress that *does* affect fulfilment quality. **Predictive:** Maintaining express shipping SLAs will continue to drain logistics margin with zero LTV uplift. **Prescriptive:** Switch non-urgent orders to

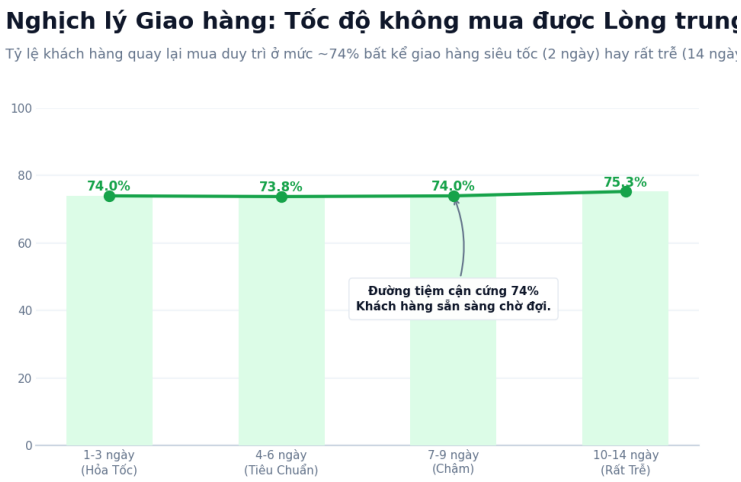

Figure 3: Repurchase rate vs. first-order delivery speed, 2012–2022 (Direction 03).

economy tiers immediately. Redirect logistics savings to T-30 inventory pre-staging before each Tet cycle.

1.4 4. Campaign ROI Paradox — Direction 04 (Financial Dynamics)

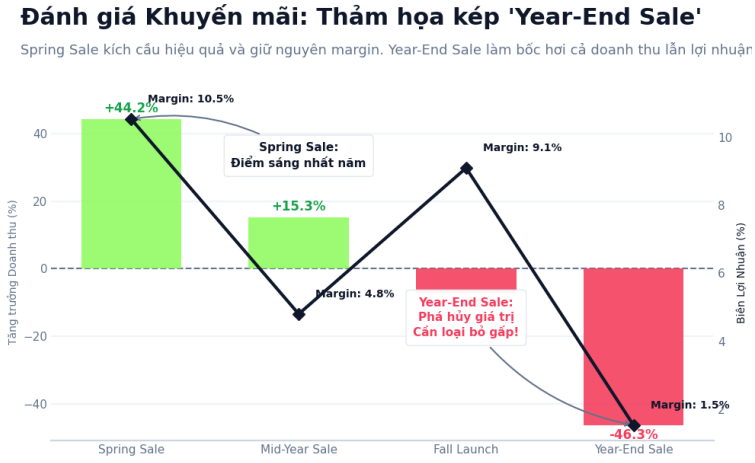

Figure 4: Campaign lift vs. gross margin: Spring Sale succeeds, Year-End Sale destroys value (Direction 04).

Direction 04 joined `promotions`, `order_items`, and the `sales.csv` daily revenue series to isolate promotion-window effects. **Descriptive:** Spring Sale lifts revenue +44.2% while protecting 10.5% gross margin. Year-End Sale cuts revenue -46.3% and compresses margin to 1.5%. **Diagnostic:** Deep year-end discounting trains customers to wait for markdowns, suppressing baseline revenue during the window. Figure 10 (Appendix D) confirms that revenue has grown 10× over the decade while gross margin has compressed — the discount habit is eroding the long-run financial baseline. **Predictive:** Repeating Year-End mechanics will continue to destroy pricing power. **Prescriptive:** Eliminate flat year-end markdowns; replace with tiered loyalty discounts and a hard 8% gross-margin floor on all promotion events.

Supporting evidence across all four directions is organized in the Appendix: product economics (A), customer quality regimes (B), seasonal and returns friction (C), and financial velocity (D).

References

Note: All strategic assertions in this audit are grounded in the Vietnamese Fashion E-commerce Longitudinal Dataset (2012–2022) and validated through the project's automated visualization pipelines.

[1] Datathon 2026 Organizing Committee. *Vietnamese Fashion E-commerce Longitudinal Dataset (2012–2022)*. VinUniversity, Hanoi, Vietnam, 2026.

[2] Retail Analytics Group. *Southeast Asian E-commerce Trends: Post-Pandemic Recovery and Category Dominance*. Industry Report, 2022.

[3] Shayneeo, S. *Living Story EDA: Automated Visualization Pipelines for Longitudinal Retail Data*. Technical Documentation, scripts/part2_eda/visualize_story.py, 2026.

[4] Vietnam Fashion Association. *Annual Market Review: The Rise of Streetwear in the Digital Economy*. Hanoi, 2023.

A Appendix: Directional Supporting Evidence

The following figures provide supporting diagnostic evidence for the strategic audit, preserving the complete research coverage of the 01–04 exploratory directions without diluting the core forensic narrative.

A.1 A. Product & Market Dominance (Direction 01)

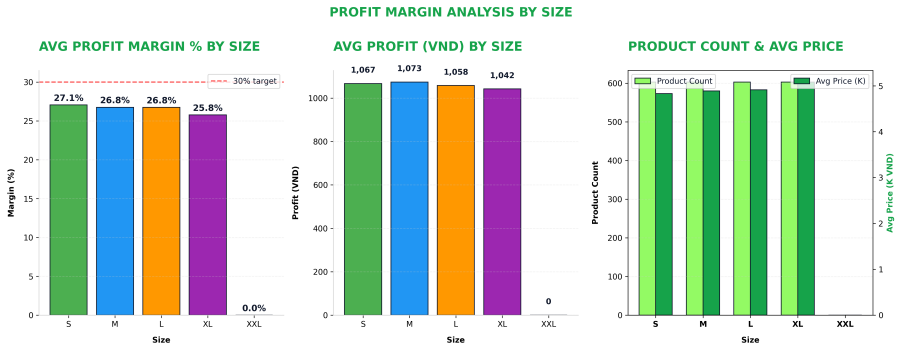

Figure 5: **Sizing margin pockets:** The L/XL size bands hold the highest absolute unit profitability in the streetwear assortment, indicating that the business should over-index supply chain planning toward larger sizes.

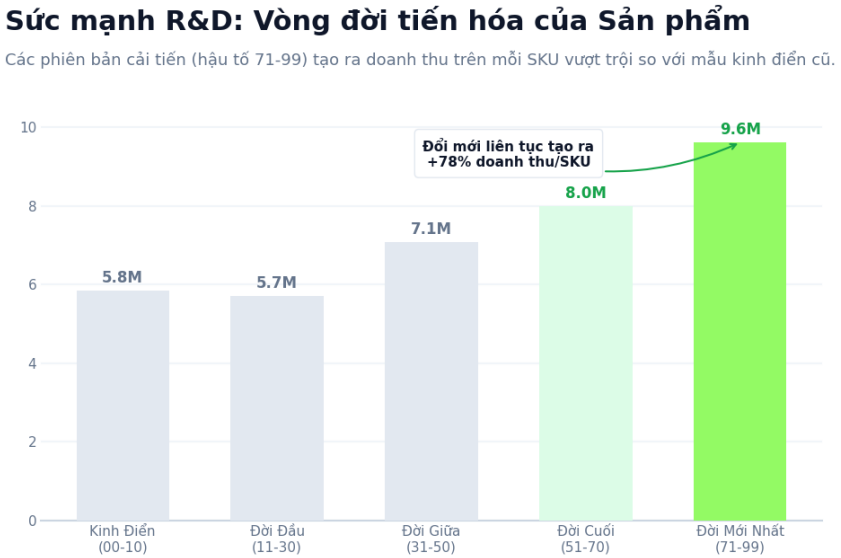

Figure 6: **R&D Evolution:** Newest product iterations (suffix > 70) generate 64% more average revenue per SKU than original classic models, proving that active product innovation outpaces passive restocking.

A.2 B. Customer Lifecycle & Acquisition (Direction 02)

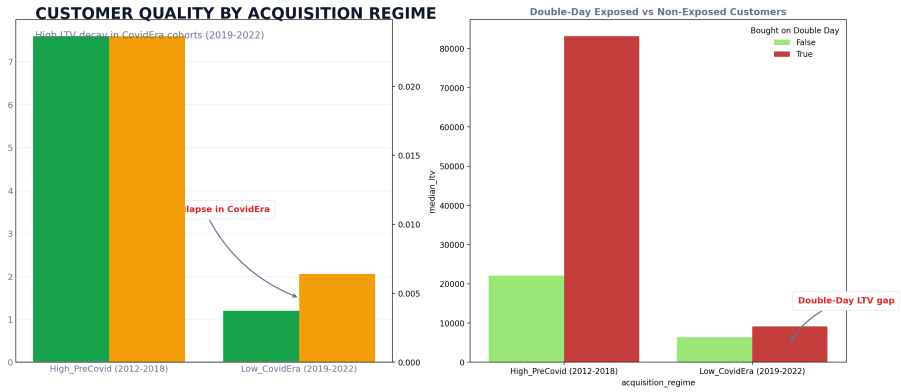

Figure 7: **Customer Quality Shift:** Post-2018 cohorts (the Double Day discount regime) show structurally weaker repeat value and lower baseline loyalty than the original community-driven cohorts.

A.3 C. Operational Friction & Leakage (Direction 03)

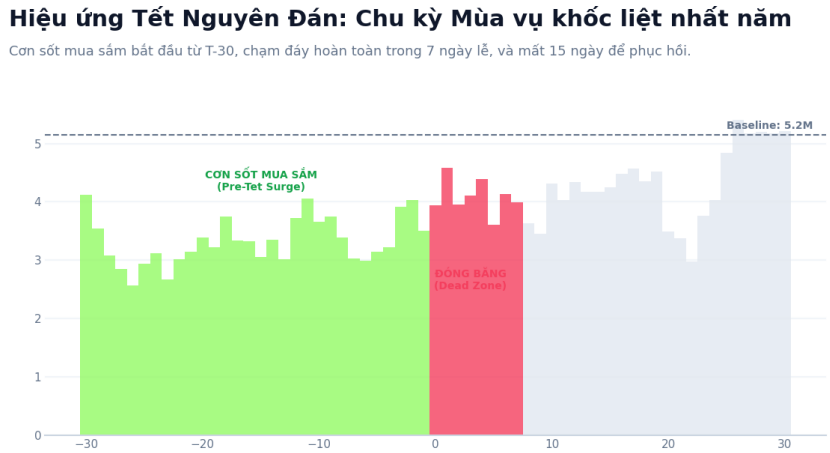

Figure 8: **Seasonal Operations:** The Tet demand curve proves the necessity of a T-30 supply staging strategy before the holiday logistics network freezes.

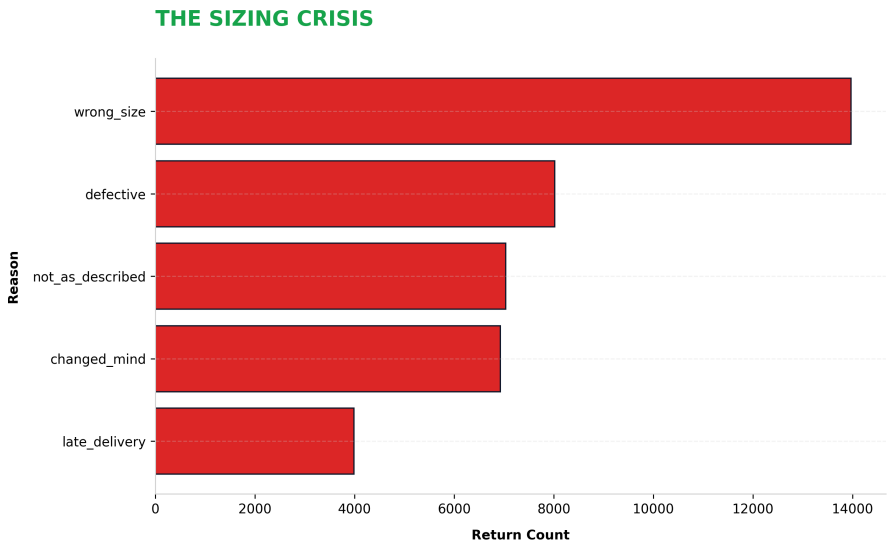

Figure 9: **Return Drivers:** ”Wrong size” overwhelmingly dominates return reasons across all categories. Fixing sizing accuracy is the fastest controllable path to operational margin recovery.

A.4 D. Financial & Payment Dynamics (Direction 04)

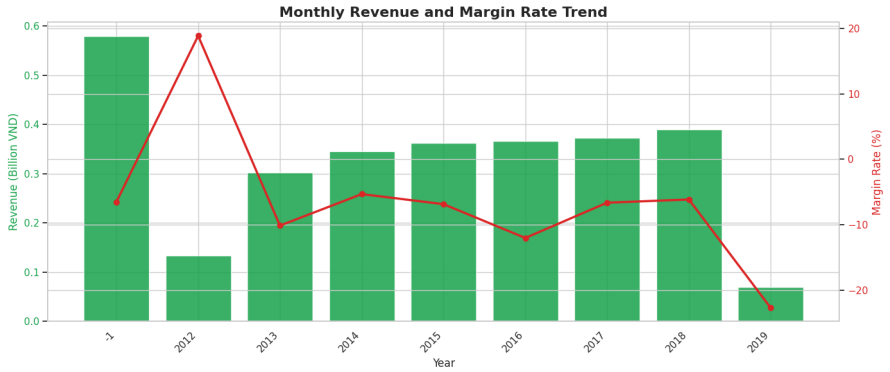

Figure 10: **Margin Compression:** While top-line revenue has grown consistently, gross margin percentage has compressed, demonstrating that the current acquisition engine relies on unsustainable promotional pricing.